Tobias Mwangi-Hartley

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PROFESSIONAL SUMMARY

Data scientist with experience developing forecasting and churn models, experimentation tools, and scheduled data workflows. Uses Python, SQL, and BigQuery to build and evaluate analytical solutions, with a focus on measurable outcomes and statistical rigour. Currently strengthening software engineering practices through pytest unit tests and seeking to develop further in production machine learning engineering.

CORE SKILLS

Machine learning: LightGBM, XGBoost, feature engineering, customer churn propensity, demand forecasting, SHAP
Programming and analytics: Python, SQL, R, pandas, scikit-learn, statsmodels
Data and workflow: BigQuery, dbt, Vertex AI Workbench scheduled notebooks, Looker
Engineering practices: pytest unit testing, Dockerfile authoring, Streamlit
Statistics: Sequential A/B testing, Bayesian hierarchical models, experimental design

PROFESSIONAL EXPERIENCE

Data Scientist | Thistlebox | 2022–Present

- Built a weekly recipe-order forecasting workflow for six weeks ahead using LightGBM and calendar, price, and menu-position features; scheduled notebook jobs write forecasts to BigQuery for operational use in Looker.
- Improved forecast WAPE from 23% to 14%; estimated food-waste reduction was 9%.
- Developed an XGBoost churn propensity model, scored weekly in BigQuery, to target retention offers to top-decile customers; a 10% holdout validation showed churn down 2.1 percentage points in the treated group.
- Built a Streamlit calculator for planning and analysing sequential A/B tests; used by about 25 people in product and growth. Wrote its Dockerfile; the data platform team handles deployment to Cloud Run.
- Started writing pytest unit tests for forecasting feature code in 2025; coverage is about 60%.
- Delivered internal “Stats for PMs” training three times.

Junior Data Analyst | Firth Energy Retail | 2020–2022

- Segmented customers and analysed tariffs using R and SQL.
- Automated monthly regulator reports with R Markdown, saving the team about two days each month.

EDUCATION

MSc in Statistics
Dissertation: Bayesian hierarchical models

ADDITIONAL PROJECT

Image classification course project | 2025

- Fine-tuned a small CNN in PyTorch on a public food images dataset as part of an online deep learning course; achieved about 80% accuracy. Learning project only.
- Tools: PyTorch, torchvision, Google Colab.